Fraud monitor

- Overview
- Patterns
- Amounts
- Detection
- Risk

type_name

- ☐ CASH_IN
- ☐ CASH_OUT
- ☐ DEBIT
- ☐ PAYMENT
- ☐ TRANSFER

Week

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

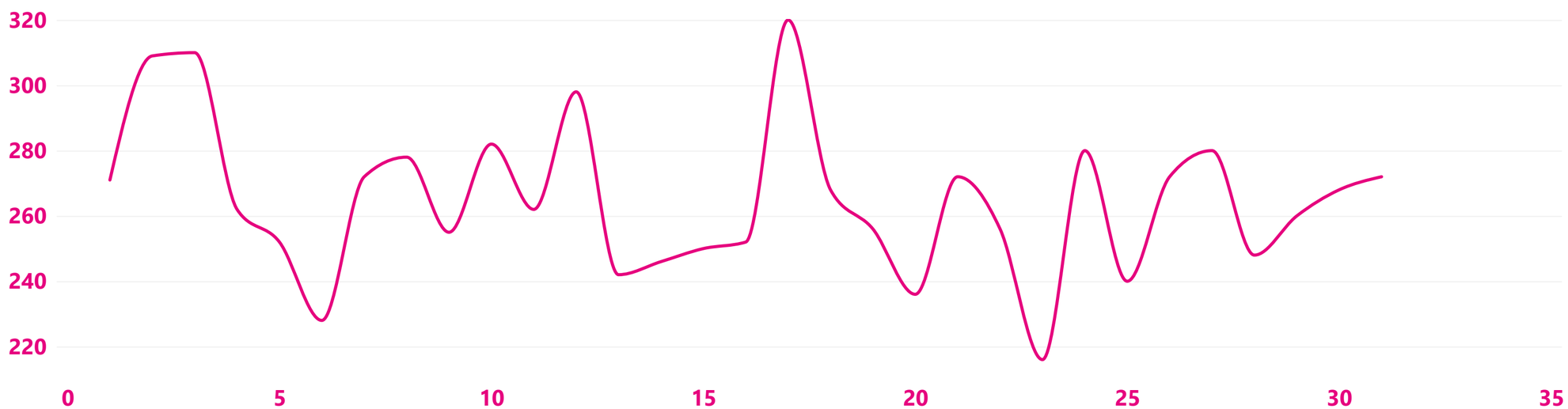
Overview

Fraud is confined to TRANSFER and CASH_OUT

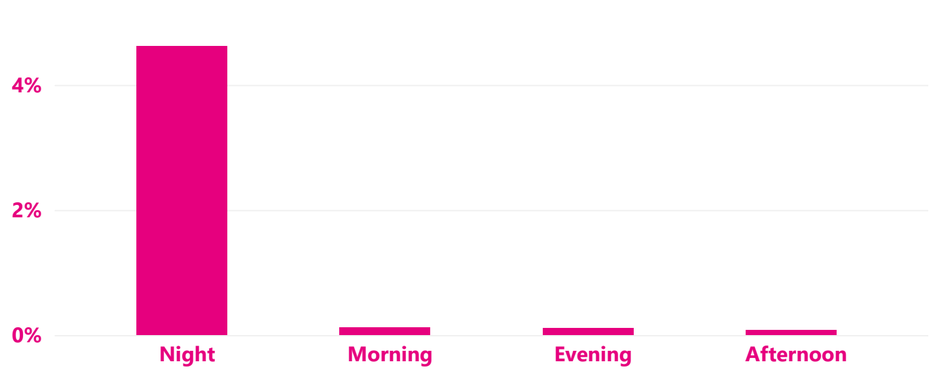
the alert system flagged only 16 of 8,213 fraud cases (0.19%).

Total Transactions	Total Amount	Fraud Amount	Fraud Count	Fraud Rate
6M	1.14T	12.06bn	8K	0.13

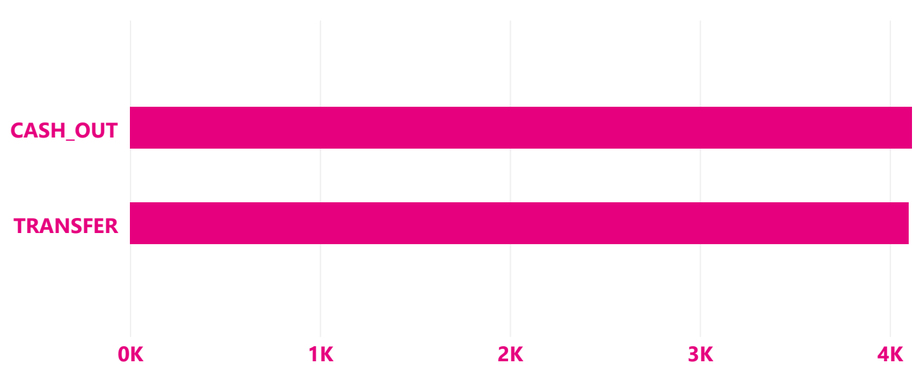
Fraud count by day



Fraud Rate % by time_of_day



fraud_count by type_name



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Patterns

Night Fraud Share
0.25

Highest-Risk Hour
4

Cash-Out Fraud Rate
0.18%

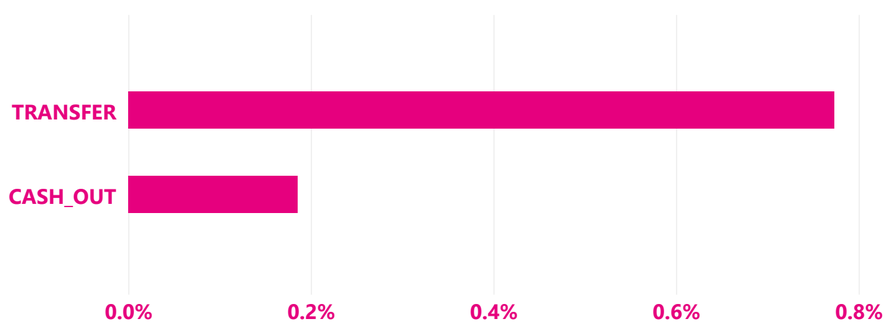
Transfer Fraud Rate
0.77%

Fraud is steady around the clock, but TRANSFER is 4x riskier per transaction

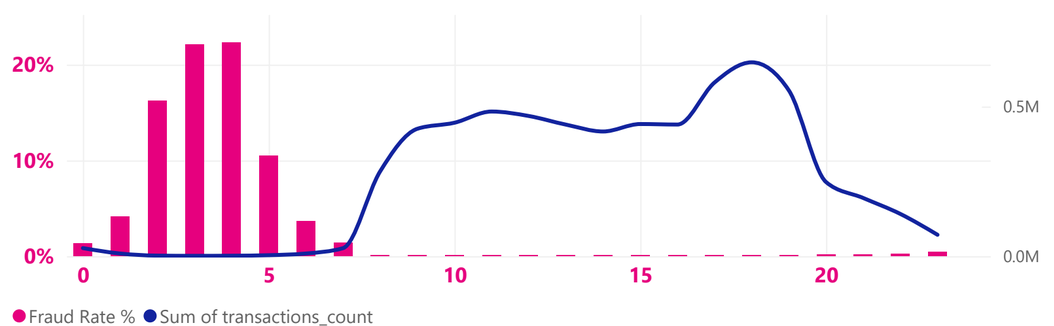
Night hours (0–5 AM) hold 25% of fraud, exactly their share of the day (6 of 24 hours), so fraud is evenly spread.

hour_of_day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	16	18	4	8	10	14	14	10	8	16	18	12	12	10	10	14	8	20	6	10	14	8	6	8	12	10	24	10	12	12	4
1	8	12	6	22	14	14	10	16	12	12	8	10	8	6	4	20	16	20	8	6	16	8	8	18	20	8	6	16	14	16	10
2	4	8	14	8	8	6	6	14	10	10	8	8	10	12	10	14	28	16	4	8	8	14	14	6	12	12	4	10	10	16	14
3	10	4	8	4	6	4	12	10	8	6	12	6	8	12	10	6	18	8	16	16	6	6	6	6	10	8	6	14	8	6	14
4	6	4	10	10	12	10	22	14	4	10	4	14	4	20	10	14	10	14	16	16	12	14	12	20	10	16	20	8	8	18	4
5	22	8	4	10	12	6	14	10	12	8	4	16	18	6	14	10	12	16	12	16	8	10	12	14	16	14	4	8	6	14	22
6	12	12	12	10	8	14	18	8	10	8	8	12	10	8	12	14	12	16	12	10	14	6	6	10	16	8	12	4	6	8	12
7	12	12	18	10	14	8	6	10	8	10	10	12	22	14	6	6	10	8	18	16	8	12	12	10	8	16	8	16	22	16	10
8	19	12	6	10	18	6	10	18	8	12	28	20	8	8	12	6	4	6	10	12	14	10	18	16	6	8	8	10	4	10	4
9	11	22	20	10	18	6	10	14	2	12	22	20	4	16	6	18	10	6	12	6	10	14	10	10	12	12	10	14	6	4	28
10	7	15	20	12	10	16	8	8	8	14	10	10	10	10	0	6	8	10	10	12	12	6	4	18	6	14	16	10	8	10	16
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16	7	12	20	6	6	12	14	16	6	12	6	18	14	6	8	10	16	28	6	10	20	14	0	14	4	16	10	10	14	8	10

Fraud Rate % by type_name



Fraud Rate % and transactions_count by hour_of_day



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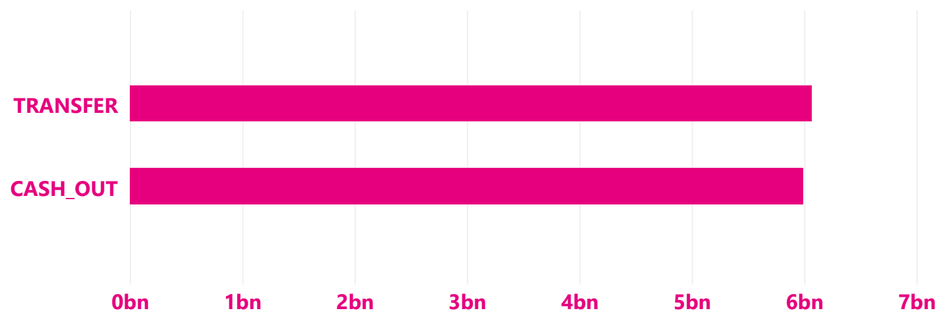
Amounts

Fraud transactions are about 8x larger than normal ones

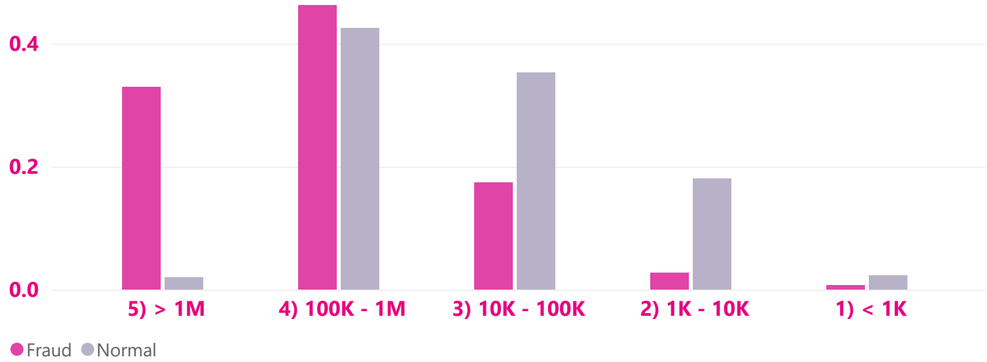
The average fraud amount is 1.47M versus 178K for normal transactions, and the largest cases hit a 10M cap

Avg Fraud Amount	Avg Normal amount	Median Fraud Amount	Fraud vs Normal Ratio
1.47M	178.20K	441.42K	8.24

Fraud amount by type

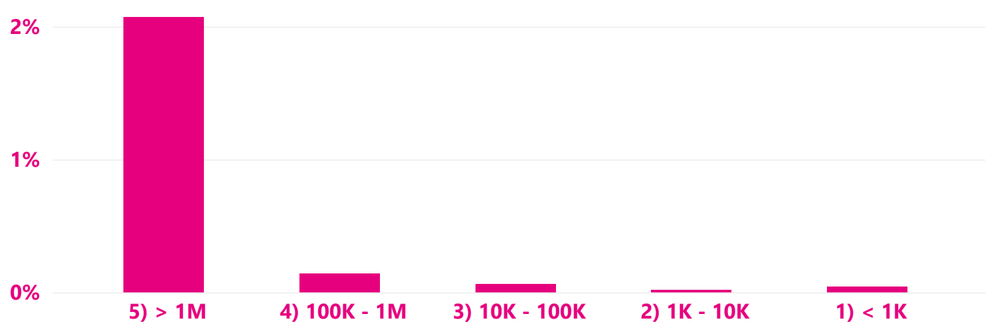


Amount mix: fraud vs normal



type_name	amount	hour_of_day	transaction_id
CASH_OUT	10,000,000.00	16	6120312
TRANSFER	10,000,000.00	8	6131081
TRANSFER	10,000,000.00	12	6152131
TRANSFER	10,000,000.00	12	6156992
CASH_OUT	10,000,000.00	15	6159419
CASH_OUT	10,000,000.00	9	6172033
TRANSFER	10,000,000.00	9	6216365
CASH_OUT	10,000,000.00	12	6314350
TRANSFER	10,000,000.00	9	6326760
CASH_OUT	10,000,000.00	16	6362206

Fraud rate by amount bucket



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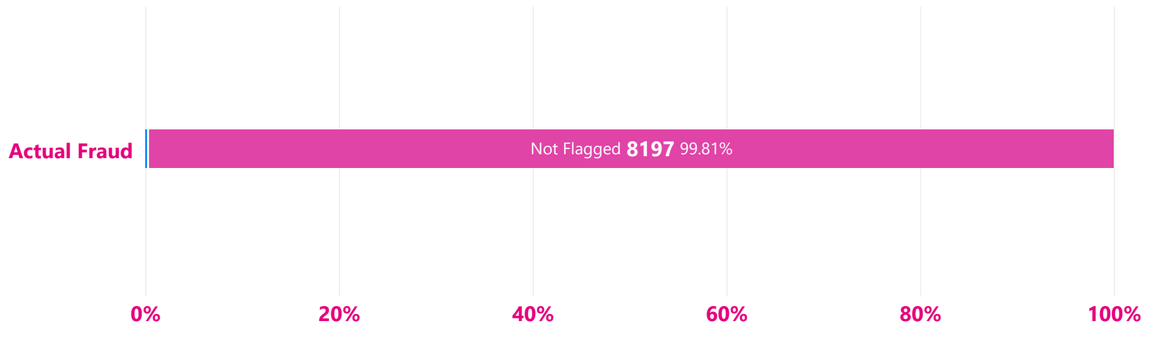
Detection

The alert system flagged only 16 of 8,213 fraud cases, a 0.19% detection rate

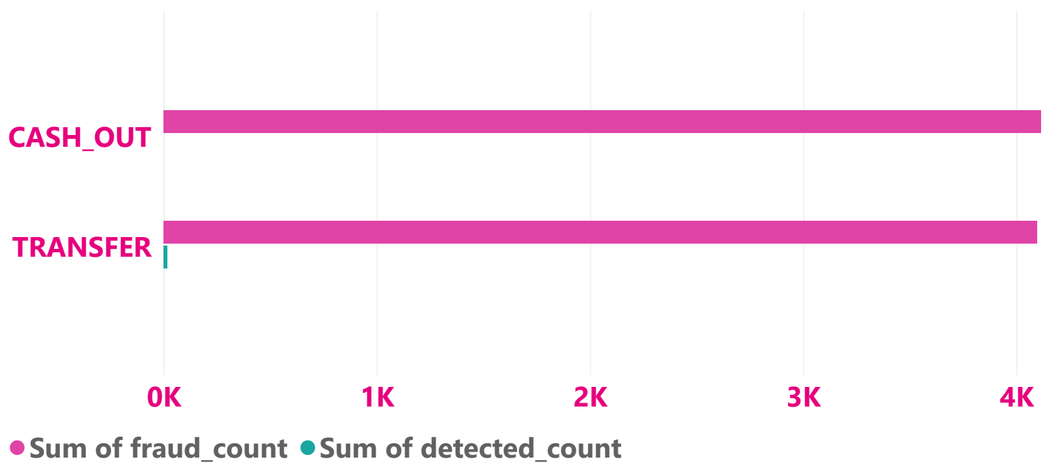
It raised no false alarms, but it never fired on CASH_OUT, which holds half of all fraud

Fraud Count	Fraud Rate	Detection Rate	Detected Fraud	Missed Fraud	False Alarms
8213	0.13	0.19	16	8197	0

Where do fraud cases go?



Fraud vs detected by type



actual_label	Flagged	Not Flagged
Actual Normal		6354407
Actual Fraud	16	8197

The 16 caught cases

type_name	amount	hour_of_day
TRANSFER	353,874.22	9
TRANSFER	399,045.08	21
TRANSFER	536,624.41	14
TRANSFER	1,343,002.08	9
TRANSFER	2,542,664.27	16
TRANSFER	3,171,085.59	5
TRANSFER	3,441,041.46	22
TRANSFER	3,576,297.10	1
TRANSFER	4,892,193.09	2
TRANSFER	4,953,893.08	19
TRANSFER	5,674,547.89	20
TRANSFER	7,316,255.05	9

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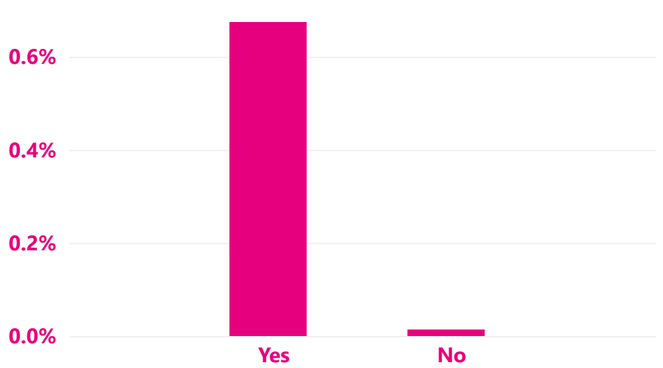
Risk

98% of fraud empties the sender's account, and no sender is ever reused

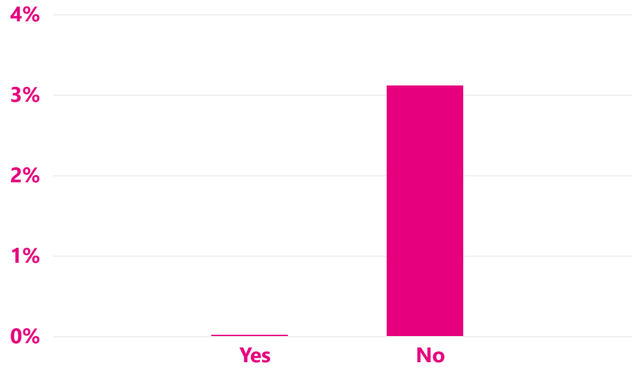
A zero balance at the destination makes fraud roughly 10x more likely, while balance mismatches are not a fraud signal at all.

Signal Fraud Rate	Emptied Share of Fraud	Risky Senders	Risky Receivers	Max Fraud per Account
0.30%	97.55%	8213	8169	1

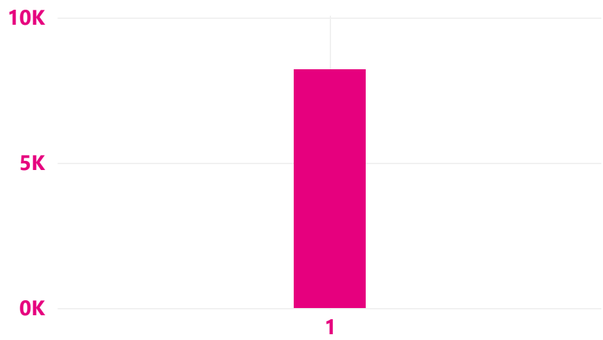
Fraud Rate by Sender Account Depletion



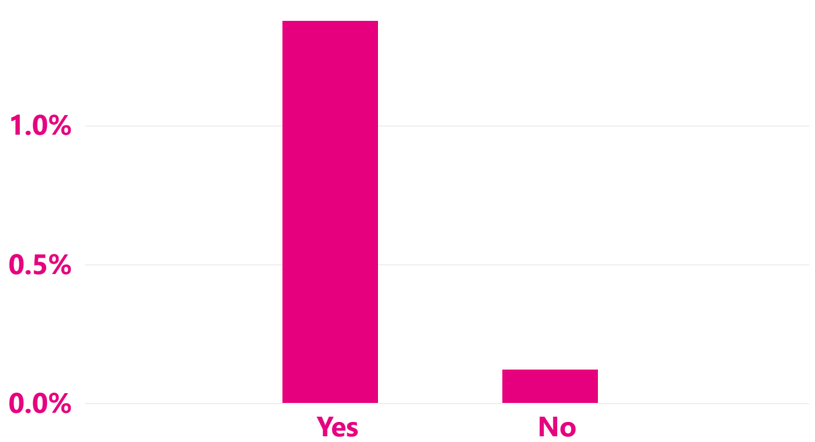
Fraud Rate by Sender Balance Mismatch



Number of Receivers by Fraud Transaction Count



Fraud Rate by Destination Zero Balance



account_number	Sum of fraud_txns_received	Sum of fraud_amount_received
C2104816494	1	10,000,000.00
C2105645265	1	10,000,000.00
C2108353776	1	10,000,000.00
C2110157840	1	10,000,000.00
C2116384638	1	10,000,000.00
C2122549663	1	10,000,000.00
C2124431731	1	10,000,000.00
C2129872070	1	10,000,000.00
C2136471075	1	10,000,000.00